

Design Document v3

Topological Proof Search: Role-Conditioned Cellular Policy Models for Lean Theorem Proving

Pre-implementation contract, revision 3

Status. Pre-implementation contract, revision 3. Items 1–10 below. Anything not in this document that emerges during implementation is added as a dated amendment, not silently changed.

Relation to v2. This is a full rewrite of v2 following the second critique exchange. See $v2 \rightarrow v3$ *changelog* below. The $v1 \rightarrow v2$ changelog is retained at the bottom for traceability.

Relation to prior paper. This document presupposes *Role-Labelled Incidence Complexes for Lean Proof States* (RLIC-1). Notation, definitions, and conditions Definitions 3.1, 3.3, 3.5, 4.5; Condition 3.6; Lemma 6.2; Theorem 6.3; Lemma 6.6; Observation 6.9 refer to that paper. The trained M5, M5a, M3, and M3+ checkpoints from RLIC-1 §7.8 are inputs to the present work; the training set of RLIC-1 is also the training set here.

Hardware envelope. Single NVIDIA RTX 4060 Ti 16 GB, LeanDojo against mathlib, no multi-GPU, no LM-from-scratch training, 4–6 calendar weeks, ≤ 250 4060-hours total.

v2 $\rightarrow$ v3 Changelog

Four amendments accepted from the second critique exchange, plus one budget reallocation that follows from them.

#	Change	v2 ref.	Why
A10	Top 2 layers of the RLIC encoder unfrozen during Mode 2 head training; lower layers frozen.	§1.6, §8.2	v2 Mode 2 head (0.4 M params on frozen 36 M encoder) faced a representational-capacity bottleneck, not a data-volume bottleneck. Partial unfreeze addresses it directly.
A11	Mode 2 SFT loss explicitly reframed as within- G_{cand} hard-negatives loss: positive is the human tactic, negatives are all other $G_{\text{cand}}(P, F(t_{\text{human}}))$ candidates.	§1.6	Free amendment. Closes part of the DPO gap relative to BFS-Prover and is the honest description of what’s happening.

#	Change	v2 ref.	Why
A12	Compiler-feedback hard-negatives training phase: on a 500-state subset disjoint from Mode 2 training, run every G_{cand} candidate through LeanDojo, record kernel verdicts, and add a margin loss with kernel-rejected candidates as hard negatives.	§1.6, §2.6	Cheap analogue of BFS-Prover’s DPO with compiler feedback (~ 8 h compute vs ~ 30 – 50 h for full DPO).
A14	G_{cand} on equality-chain proof states filters out <code>rw [h]</code> candidates whose LHS does not textually appear in the current goal (and analogously for <code>rw [← h]</code>). 3.6-compliant syntactic check.	§1.4, §7.4	v2 proof of Theorem 7.4(a) had a flaw: rewrites whose terms don’t appear in the goal get kernel-rejected, consuming expansions and breaking $O(n)$ to $O(n^2)$. Textual-presence filter restores $O(n)$ cleanly.
B1	Held-out mathlib slice filter tightened: proof-length ≤ 15 (was ≤ 30); additionally require the human proof’s tactics lie within G_{cand} ’s expressive range under B2.	§5.1	v2 slice contained proofs whose human solution G_{cand} structurally cannot reproduce, capping coverage at $< 50\%$ on topology/calculus prefixes.
B2	G_{cand} for the <code>apply</code> family augmented with a 3.6-compliant retrieval shortlist: BM25 over premise statements from a frozen mathlib snapshot, top- $K = 8$ per state. Retrieval pool excludes the held-out theorems.	§1.3	Without retrieval, coverage on mathlib non-local proofs is catastrophic. BM25 with a frozen index satisfies Condition 3.6 and restores coverage while preserving leakage-safety.
R1	Diagnostic miniF2F reduced from 244 to 100 theorems (random sample, seed 17).	§5.2, §10	Reallocates ~ 20 h to fund A10 + A12 + B2 within the 250-h ceiling. Diagnostic role doesn’t need 244.

Things that did *not* change from v2: Mode 1 / Mode 2 two-stage policy structure (C1 of v2); value head dropped from priority (C2); length-normalised priority $f(P) = \log \pi_{\text{path}}(P)/d(P)^{\alpha_{\text{len}}}$ (C5); the M3+ control as the grading-isolating ablation; the three-seed protocol on M5 and M3 (C7); the IHES qualitative protocol; the search-soundness theorem and budget-monotone lemma.

1 Formal setup of the search-coupled system

1.1 The policy: Mode 1 and Mode 2

Mode 1 (family classification). Inherited from RLIC-1:

$$\pi_1(\text{family} \mid K_{\text{afford}}(P)) = \text{softmax}(W_1 \cdot \text{pool}_{\text{class}}(h^L(K_{\text{afford}}(P)))),$$

over $\mathcal{F} = \{\text{rw}, \text{apply}, \text{exact}, \text{induction}, \text{constructor}, \text{simp}, \text{intro}, \text{rfl}, \text{ring}\}$. Input is $K_{\text{afford}}(P)$. The family-leakage-safety contract of RLIC-1 holds.

Mode 2 (candidate-instance scoring). A small head trained as part of this work (§1.6) that scores candidates within a chosen family:

$$\pi_2(a \mid K_a(P, A)) = \text{softmax}_{a \in A}(w_2^\top \text{MLP}(h_{c_a}^L)),$$

with $A = G_{\text{cand}}(P, F)$. Mode 2 satisfies Condition 3.6 because the candidate set A is generated by a family-universal procedure (§1.3) and the candidate-action cells in $K_a(P, A)$ carry only the structural signature of a , not a target-instance label.

Joint policy. Per-step probability of taking tactic a :

$$p(a \mid P) = \pi_1(F(a) \mid K_{\text{afford}}(P)) \cdot \pi_2(a \mid K_a(P, A(F(a)))),$$

where $A(F(a)) = G_{\text{cand}}(P, F(a))$ includes the retrieval shortlist of B2 where applicable. Path log-probability is the sum of per-step log-probabilities.

1.2 The value head: present but not used

$V(P)$ from RLIC-1 remains computable but is not in the priority function. Calibration of V on real search-failure states (mined during M5+M2 search) is reported as a diagnostic for the value-head consistency question; $\beta = 0$ is the headline.

1.3 The candidate generator G_{cand}

$G_{\text{cand}} : (P, \text{family}) \rightarrow \text{list}[\text{Tactic}]$ is deterministic in (P, Π) and uses only Condition 3.6(ii) queries. Per-family specs:

- **rfl:** `[rfl]`.
- **ring:** `[ring]` if the ambient type has a `CommRing/Ring/CommSemiring` instance at depth ≤ 2 , else `[]`.
- **intro:** `[intro x_canon]` where `x_canon` is a canonical name.
- **simp:** `[simp, simp [*], simp_all, simp only []]`. **No retrieval here in v3;** retrieval-augmented `simp` is a known follow-up.
- **constructor:** from `constructor-opp` cells. Up to six candidates.
- **induction:** from `induction-opp` cells. Up to 16 candidates.

- **rw**: from **rewrite-opp** cells, capped at the most-recent 12 equality hypotheses. **Equality-chain subcase**: §1.4 (Fiedler ordering) + A14 textual-presence filter.
- **apply**: from **apply-opp** cells (local hypotheses with $\text{head}(\text{type}(h)) = \text{head}(\text{goal})$ via first-order syntactic matching at Θ); **augmented in v3 (B2) with up to 8 retrieved external lemmas** (§1.5). Capped at the most-recent 16 local + 8 retrieved = 24 candidates max.
- **exact**: from **apply-opp** cells with exact type match, plus **exact h.symm** for equality symmetry. Capped at 8.

Per-family fan-out cap. Default 24 candidates per family. Total per state $\leq \sim 100$ (slight increase over v2 due to B2 retrieval).

Coverage rate. Reported up-front: the fraction of (state, human-tactic) pairs in the RLIC-1 training set for which the human tactic is in $G_{\text{cand}}(P, \text{family}(t_{\text{human}}))$, where G_{cand} now includes B2 retrieval. This is the Mode 2 upper-bound accuracy.

1.4 Equality-chain subcase: spectral candidate ordering plus textual filter (A14)

Detection. Same as v2: a proof state P is an *equality-chain state* if every local hypothesis has type $a_i = a_{i+1}$ for a chain in some Lean type T , and the goal is $a_1 = a_n$. Syntactic check using only Condition 3.6(ii)(a).

Spectral ordering. Build the cellular sheaf $\mathcal{F}$ of §7.4 on the chain; compute sheaf Laplacian $L = \delta^* \delta$; compute the Fiedler vector v_1 for λ_1 . The *spectral score* of h_i is $\sigma(h_i) = |v_1(a_{i+1}) - v_1(a_i)|$. Candidates ordered by descending σ .

Textual-presence filter (A14, new in v3). Before submitting candidates to the priority queue, G_{cand} discards:

- **rw [h_i]** if neither $\text{lhs}(h_i) = a_i$ nor $\text{rhs}(h_i) = a_{i+1}$ appears as a subterm of the current goal at transparency Θ ;
- **rw [$\leftarrow$ h_i]** if neither a_{i+1} nor a_i appears in the current goal.

The check is syntactic at transparency Θ equivalent to “would Lean’s **rw** tactic *pattern-match* against the goal?” and does not execute the rewrite. 3.6-compliant: parse-tree query plus kernel-typing query.

Why A14 matters for Theorem 7.4(a). Without A14, the Fiedler ordering places middle-of-the-chain rewrites at high priority, but these rewrites’ LHS/RHS terms do not appear in the initial goal $a_1 = a_n$; the kernel rejects them, expansions are consumed, and the $O(n)$ bound of Theorem 7.4(a) breaks to $O(n^2)$. With A14, only rewrites whose terms are textually present in the current goal pass the filter; at each search state of the chain, at most two candidates pass (**rw [h_left]** and **rw [$\leftarrow$ h_right]**), and the Fiedler ordering distinguishes them. The $O(n)$ bound is restored as a theorem about the actually implemented system.

Fallback. If P is not an equality-chain state, the default G_{cand} ordering applies (Mode 2 re-scores within family in any case).

1.5 Retrieval shortlist (B2, new in v3)

Goal. Extend G_{cand} ’s `apply` family with candidate lemmas from outside the local context, while preserving Condition 3.6.

Index construction (one-time, pre-training). Take a snapshot of mathlib at a fixed commit c_{ref} chosen before any experiment runs. Extract all theorem and lemma statements at c_{ref} . Build a BM25 index over the pretty-printed statements (tokenised by standard whitespace + identifier split; standard BM25 parameters $b = 0.75, k_1 = 1.2$). The index is frozen at the start of the study and is not updated.

Important exclusion. The held-out mathlib slice’s theorems are *removed* from the retrieval pool. This prevents the trivial leakage of retrieving “the lemma I am trying to prove” as an `apply` candidate. The exclusion set is computed deterministically from the held-out slice’s file paths.

Per-state lookup. At any proof state P , the query is the pretty-printed goal text. BM25 returns the top- $K = 8$ premise statements by score. These are added to $G_{\text{cand}}(P, \text{apply})$ as `apply lemma_name` candidates (with `lemma_name` the retrieved lemma’s fully-qualified name).

3.6-compliance argument.

- (i) **Determinism:** BM25 with a fixed index, fixed tokenisation, fixed K is deterministic in the query, which is deterministic in P . ✓
- (ii) **Computational boundary:** the query is constructed from the parse tree; the index is precomputed and read-only; no kernel reduction, no tactic execution. ✓
- (iii) **Independence of next move:** the premise pool is the entire c_{ref} snapshot minus the held-out slice, fixed at study start. ✓

Therefore by Lemma 6.2 of RLIC-1 (compositional no-leakage), G_{cand} augmented with B2 retrieval continues to satisfy Condition 3.6.

Caveat. BM25 is sparse and surface-level. It will produce many false-positive candidates. The Mode 2 head must filter these during scoring. This shifts work from candidate generation to candidate scoring; if Mode 2 lacks the capacity to filter false retrievals, B2 hurts more than it helps. Checked empirically; the no-retrieval G_{cand} is also reported in §7.

1.6 Mode 2 head: training procedure (A10, A11, A12)

Phase 0: data preparation. For each (P, t_{human}) in the RLIC-1 training set: set $F = \text{family}(t_{\text{human}})$; generate $A = G_{\text{cand}}(P, F)$ (including B2 retrieval). If $t_{\text{human}} \in A$, the pair is included in Mode 2 training. If $t_{\text{human}} \notin A$, the pair is excluded and counted toward the coverage-rate statistic.

Phase 1: hard-negatives SFT (A11). For each included (P, t_{human}, A) :

- Positive: t_{human} .
- Negatives: all other candidates in A (hard-negative set within the same family).

- Loss: cross-entropy over the softmax of Mode 2 scores within A .

Functionally identical to v2’s Mode 2 SFT, reframed as a hard-negatives loss with within-family competitors as negatives.

A10: partial encoder unfreeze. During Phase 1, the top 2 layers of the RLIC encoder are unfrozen and fine-tuned jointly with the Mode 2 head. The bottom $(L - 2) = 2$ layers remain frozen. Optimiser: AdamW with $\eta = 3 \times 10^{-4}$ for the head, $\eta = 1 \times 10^{-4}$ for the unfrozen encoder layers. Cosine schedule with 400 warmup steps.

Phase 1 training schedule. 40 000 optimiser steps, batch 32, BF16. Estimated 6 h per architecture (M5, M3, M3+) on a 4060 (up from v2’s 4 h due to encoder partial-unfreeze). Three architectures = 18 h.

Phase 2: compiler-feedback hard negatives (A12, new in v3). After Phase 1 converges, run an additional 10 000-step training pass with the following data.

Data collection. Choose 500 states P from the validation split (disjoint from Phase 1 training). For each P , for each family F , for each candidate $a \in G_{\text{cand}}(P, F)$ (across all families including B2 retrieval): submit `LeanDojo.run_tac(P, a)` and record the kernel verdict (accept / reject / timeout). The data collection is amortised across all three architectures (M5, M3, M3+) since kernel outcomes depend only on (P, a) , not on which RLIC encoder reads them.

Estimated cost of data collection: $500 \times \sim 25 \times \sim 1.2 \text{ s} \approx 4 \text{ h}$, done once and reused.

Phase 2 loss. For each (P, F) pair, partition $A = G_{\text{cand}}(P, F)$ into:

- A_{acc} : kernel-accepted candidates.
- A_{rej} : kernel-rejected candidates.
- A_{timeout} : timeout (treated as rejected for ranking purposes).

Add a margin loss term:

$$\mathcal{L}_{\text{margin}}(P, F) = \mathbb{E}_{a^+ \in A_{\text{acc}}} \left[\mathbb{E}_{a^- \in A_{\text{rej}}} [\max(0, m - (\pi_2(a^+) - \pi_2(a^-)))] \right],$$

with margin $m = 0.5$. Combined loss = Phase 1 cross-entropy + $\lambda \cdot \mathcal{L}_{\text{margin}}$ with $\lambda = 0.3$.

Estimated Phase 2 training cost: $\sim 1.5 \text{ h}$ per architecture $\times 3 = 4.5 \text{ h}$. Plus 4 h data collection (amortised). Total A12 cost: $\sim 8.5 \text{ h}$.

Why this is the right amount of DPO-substitute. BFS-Prover ran DPO over millions of state-tactic pairs with full RL infrastructure. We are doing a cheap, targeted analogue: 500 states, kernel-feedback negatives, margin loss. It will not match BFS-Prover’s policy quality, but it will give us *some* negative-signal training, addressing the most concrete gap identified in the second critique. The full DPO-with-compiler-feedback pipeline remains future work; we flag this explicitly in §10.

Held-out validation. A 1000-state validation split (disjoint from Phase 1 and Phase 2 training) is used for early-stopping and reporting Mode 2 top-1 within-family accuracy.

Parameter count. Mode 2 head: ~ 0.4 M params. Unfrozen RLIC encoder layers: ~ 6 M params per architecture. Total *trainable* parameter count during Mode 2 training: ~ 6.4 M per architecture. Total model parameter count (frozen + trainable): unchanged at ~ 36 M.

1.7 Hybrid candidate generation with an LM (optional, leakage-flagged)

The hybrid

$$G_{\text{cand}}^{\text{hyb}}(P, F) = G_{\text{cand}}(P, F) \cup G_{\text{lm}}(P, F)$$

is reported only as a side experiment. With B2 retrieval already providing leakage-safe external lemmas, we expect the hybrid column’s marginal advantage to be smaller than it would have been in v2. The leakage discipline holds at the policy-head level; not at the combined-system level when hybrid is enabled. The main tables in §7 separate **leakage-safe** (G_{cand} with B2) and **hybrid** columns.

1.8 The state-action pair the search consumes

At each search node: $(P, K_{\text{afford}}(P), K_a(P, A_{\text{full}}), \pi_1(P), \pi_2(P), \text{parent}, \text{depth}, \text{path})$, where $A_{\text{full}} = \bigcup_F G_{\text{cand}}(P, F)$ includes the B2 retrieval shortlist. K_{afford} and K_a computed once on first visit; π_1 and π_2 computed for the top- k families and cached.

2 The search procedure

2.1 Best-first search over proof states

Open list: priority queue keyed by $f(P)$ (§2.2). Closed list: state-hash set. Expansion order: highest priority first. Deterministic tie-breaking: FIFO on insertion, then lexicographic on tactic string.

2.2 The priority function

$$f(P) = \frac{\log \pi_{\text{path}}(P)}{d(P)^{\alpha_{\text{len}}}}$$

$\log \pi_{\text{path}}(P)$ = sum of per-step joint Mode-1 $\times$ Mode-2 log-probs along the path from root to P . $d(P)$ = depth. $\alpha_{\text{len}} \in [0, 1]$ swept on a 50-theorem dev split; BFS-Prover’s reported sweet spot is $\alpha_{\text{len}} \approx 0.5$.

2.3 Expansion procedure

At node P :

1. Compute $\pi_1(\text{family} \mid K_{\text{afford}}(P))$.
2. Take top- k_{fam} families (default 5).
3. For each: build $A_F = G_{\text{cand}}(P, F)$ (with B2 retrieval and A14 filter on equality chains); build $K_a(P, A_F)$; compute $\pi_2(a \mid K_a(P, A_F))$.
4. Concatenate $\{(a, \pi_1(F) \cdot \pi_2(a \mid \cdot))\}$ sorted by joint prob.
5. Take top- $k_{\text{exp}} = 8$.

6. For each a : `LeanDojo.run_tac(P, a)`. Push valid successors with their $f(P')$. On `no_goals`, terminate with success.

2.4 Termination

- **Success.** A successor has `no_goals`. Replay through Lean (Theorem 7.1).
- **Budget exhausted.** Total kernel calls reach B .
- **Per-theorem timeout.** 5 min wall.

2.5 Budget unit

One expansion = one `LeanDojo.run_tac` call. Standard B-grid: $B \in \{16, 32, 64, 128, 256\}$.

2.6 Why BFS, not MCTS

- (i) BFS empirically wins at this scale (BFS-Prover, ACL 2025) with length normalisation and no value head, exactly the algorithmic template we follow.
- (ii) MCTS rollouts are expensive on a single 4060.

We acknowledge BFS-Prover’s success required DPO with compiler feedback, which we approximate via A12 (§1.6). Our A12 is a cheap version; matching BFS-Prover’s full DPO is out of budget and out of scope.

3 The principal hypothesis

3.1 Statement

H_principal (end-to-end search-coupled, v3). At fixed search budget B , at compute parity in both training and search, and using the leakage-safe G_{cand} (with B2 retrieval and A14 filter, no $G_{\text{cand}}^{\text{hyb}}$):

M5 + Mode 2 + best-first solves strictly more theorems on the held-out mathlib slice than M3 + Mode 2 + best-first, and the absolute pass-rate gap grows monotonically with B over $B \in \{16, 32, 64, 128, 256\}$.

Formally, with $\text{pass}(\text{System}, B)$ the pass rate on the held-out mathlib slice (post-B1 filter):

- (i) $\text{pass}(\text{M5}+\text{M2}, B) > \text{pass}(\text{M3}+\text{M2}, B)$ for every B in the tested range;
- (ii) $\Delta(B) = \text{pass}(\text{M5}+\text{M2}, B) - \text{pass}(\text{M3}+\text{M2}, B)$ is non-decreasing in B within noise.

3.2 Distinguished from weaker hypotheses

- **H_a** (Mode 1 top-1 of $\text{M5} > \text{M3}$): established by RLIC-1. Re-confirmed as sanity check.
- **H_b** (V-head calibration of $\text{M5} > \text{M3}$): tangential; V not in priority function.
- **H_c** (Mode 2 top-1 of $\text{M5} > \text{M3}$): not established by RLIC-1; new test for v3.

H_principal is the search-coupling contribution. It is not implied by H_a or H_c alone.

3.3 Secondary: H_diag on miniF2F-100

On the 100-theorem miniF2F-test diagnostic subset (R1), M5+M2 solves at least $1.5\times$ as many theorems as the no-policy baseline at $B = 256$. We do *not* expect miniF2F pass rates competitive with frontier provers (BFS-Prover 72.95%, DeepSeek-Prover-V2 88.9%, HILBERT 99.2%); H_diag is a much weaker functional verification.

3.4 Pre-registered statistical procedure

Paired bootstrap CIs (10^4 resamples), signed permutation tests with Bonferroni at $\alpha = 0.05$, Spearman ρ between B and $\Delta(B)$, Cohen’s h at highest B . Three seeds on M5+M2 and M3+M2; one seed on M3++M2, LM, and Mode-1-only ablation.

4 Baselines

4.1 Six systems

- (i) **M5 + Mode 2 + best-first.** Three seeds.
- (ii) **M3 + Mode 2 + best-first.** Parameter-matched (35.81 M). Three seeds.
- (iii) **M3+ + Mode 2 + best-first.** Grading-isolating control (35.92 M). One seed.
- (iv) **Small LM + best-first.** Pythia-160M LoRA ($r = 16$), 80 000-step SFT on the RLIC-1 training set. Acts as candidate generator + scorer in tandem. One seed.
- (v) **No-policy floor.** Fixed family order, G_{cand} within. One run.
- (vi) **M5 + Mode-1-only.** Ablation isolating Mode 2 contribution. Reported on miniF2F-100 only.

4.2 Parameter and compute parity

M3 and M3+ parameter-matched to M5 within $\pm 1\%$ at the encoder level. Mode 2 heads are identical architecture across M5/M3/M3+. All RLIC encoders inherit 80 000-step training from RLIC-1; Mode 2 heads get 40 000 + 10 000 (Phase 1 + Phase 2) steps each.

The LM gets 80 000 SFT steps. The LM is *not* parameter-matched to M5 (it is a different model class); we document this honestly.

4.3 Reporting

For each system $\times$ seed $\times$ benchmark: pass-rate at each B with 95% CI; Mode 1 and Mode 2 top-1 on validation; G_{cand} coverage rate (with B2 retrieval breakdown); mean expansions per success / failure; per-slice breakdown (rewrite-heavy, multi-arg-app, induction, residual, **equality-chain** (new under A14)).

5 Evaluation set

5.1 Primary: held-out mathlib slice (B1, B2)

File-path prefixes (disjoint from RLIC-1 train): Mathlib/Topology/, Mathlib/CategoryTheory/, Mathlib/Analysis/Calculus/, Mathlib/Combinatorics/.

Proof-length filter (B1): $5 \leq \text{proof_length} \leq 15$ (was ≤ 30 in v2). Tighter cap.

B1 secondary filter. The human proof’s tactics must lie within G_{cand} ’s expressive range, defined as: every tactic in the proof is either a local tactic (`rfl`, `simp`, `intro`, etc.) or an `apply lemma / rw [lemma]` where `lemma` is either local or appears in the B2 retrieval top- $K = 16$ of any state along the proof. This filter is computed by running $G_{\text{cand}} + \text{B2}$ on each state of the human proof and checking inclusion. States whose human tactic isn’t covered are reported as *coverage misses*; theorems with any miss are excluded from the slice. The miss rate becomes a property of the slice we constructed, reported alongside.

State budget. Initial proof state must fit within parser budgets $(|K_0|, |K_1|, |K_{\geq 2}|) = (256, 512, 64)$.

Deduplication. Hash each theorem statement after canonical form; exclude collisions with RLIC-1 train.

Retrieval-pool exclusion (B2). Theorems in this slice are removed from the BM25 retrieval index before any experiment runs.

Target size. 500 theorems. If filter yields more, take a fixed seed-17 random sample. If fewer, accept $300 \leq N \leq 500$ and document the actual N .

What this means for H_principal. The slice is now explicitly *constructed* to be within G_{cand} ’s reach. This is a methodological strength (we are not asking the system to do something it structurally cannot) but also a constraint on the contribution’s reach: we are evaluating compute-parity inductive bias on solvable-by- G_{cand} problems, not on arbitrary mathlib.

5.2 Diagnostic: miniF2F-test, 100-theorem subset (R1)

100 theorems sampled from miniF2F-test with seed 17 (fixed). Reported at $B = 256$ with all systems including the Mode-1-only ablation. Functional verification only; no SOTA chase.

5.3 Deferred: novel_premises split

Same as v1/v2. Not exercised.

5.4 No-leakage spot-check

1000 states from each set, Algorithm 4 probes, $\geq 99\%$ pass-rate required.

6 The qualitative-evidence protocol (IHES card)

Pre-committed.

6.1 Structural-complexity score (unchanged from v2)

$S(T) = z(s_1(T)) + z(s_2(T)) + z(s_3(T))$ where s_1 = grade-2 cell count, s_2 = longest rewrite-chain, s_3 = hypothesis-graph diameter, all computed on the initial state of T .

6.2 Selection (updated for R1)

1. Compute $S(T)$ for all T in miniF2F-100.
2. Top decile (10 theorems).
3. Sub-select those where the LM baseline fails at $B = 256$.
4. Cap at 10.

6.3 Qualitative table

Statement; $S(T)$; M5+M2 trace; M3+M2 trace; LM trace; mathematical annotation. 2–4 sentence annotation column is the IHES payload.

6.4 Falsifiability

If M5+M2 fails on a majority of high- $S(T)$ selected theorems, the structural-mechanism prediction is unsupported; we report this honestly.

7 Mathematical content

7.1 Search-soundness theorem

Theorem 1 (Search-soundness). *Any proof returned by the §2 search with policy (π_1, π_2) , candidate generator G_{cand} (with B2 retrieval and A14 filter), and Lean kernel verification is accepted by Lean. The neural components never enter the trusted base.*

Proof. Induction on path length; kernel acceptance is the verification step. □

7.2 Budget-monotone improvement lemma

Lemma 2 (Budget-monotone improvement). *For fixed (π_1, π_2, G, f) and deterministic tie-breaking, $\text{pass}(B') \geq \text{pass}(B)$ for $B' > B$.*

Proof. Deterministic BFS extends its expansion prefix monotonically. □

7.3 Value-head consistency: condensed (deferred)

V calibration on real failure states is reported as a diagnostic; retraining V on mined failures is future work. The “conditions under which V would generalise” (formerly v1 Definition 7.3) are listed in a single paragraph and deferred.

7.4 Cellular-sheaf variant for equality-chain proof states (reformulated under A14)

Setting. Equality-chain state P : hypotheses $h_1 : a_1 = a_2, \dots, h_{n-1} : a_{n-1} = a_n$; goal $a_1 = a_n$; all in a common Lean type T . The proof-state-cell-complex subcase of RLIC-1 Definition 3.3.

Sheaf construction. As in v2: stalks $\mathcal{F}(a_i) = \mathbb{R}^d$; restriction maps along **source** and **target** incidences of each **eq-lit** cell are identity with opposite orientation; parser-imposed path-identifications make the assignment into a true sheaf (not just a quiver representation, per RLIC-1 Remark 4.3).

Sheaf Laplacian. $L = \delta^* \delta$ on 0-cochains over $\{a_1, \dots, a_n\}$. Tridiagonal, weighted-path-graph form. Eigenvalues $0 = \lambda_0 < \lambda_1 \leq \dots \leq \lambda_{n-1}$. The Fiedler vector v_1 is used by G_{cand} to order **rw** candidates (§1.4) after the A14 textual-presence filter has restricted the candidate set to terms textually present in the current goal.

Theorem 3 (Reformulated under A14). *Let P be an equality-chain proof state of length n , and let L be its sheaf Laplacian. Let G_{cand} use the A14 textual-presence filter together with the Fiedler ordering (§1.4). Then:*

- (a) *Under the length-normalised best-first search of §2 with $\alpha_{\text{len}} = 0.5$ and the joint policy of §1.1, search terminates with a proof in at most n expansions on any equality-chain state where $\lambda_1(L) > 0$.*
- (b) *When the chain contains structurally redundant hypotheses (e.g., h_i and h_j with overlapping rewrite scopes), the Fiedler vector’s contrast across the redundant edges drives the discriminative ordering; the number of expansions until the first success is bounded by $O(n + r)$ where r is the number of redundancies.*

Proof. (a) At any search state on the chain, the A14 filter restricts the candidate set to rewrites whose terms are textually present in the current goal.

Base: the initial state has goal $a_1 = a_n$. The textual filter passes exactly two candidates among the **rw** family: **rw** [h_1] (LHS a_1 is in goal) and **rw** [$\leftarrow$ h_n-1] (RHS a_n is in goal); all other **rw** [h_i] and **rw** [$\leftarrow$ h_i] have no term in common with the goal and are filtered out. Mode 2 picks one (via the joint policy); Fiedler tie-breaks. Both options reduce the chain length by 1. One expansion consumed.

Inductive step: after k expansions the goal is of the form $a_p = a_q$ with $q - p < n$ (chain length strictly decreasing each step). The A14 filter again admits at most two candidates. One expansion reduces chain length by 1.

Termination: after at most $n - 1$ chain-shortening expansions, the goal is $a_p = a_p$ for some p ; the next expansion tries **rf1** (in $G_{\text{cand}}(P, \text{rf1}) = [\text{rf1}]$, always passing the A14 filter trivially); it succeeds. Total: at most n expansions.

(b) When there are r redundant hypotheses, the textual filter may pass more than two candidates at some states. The Fiedler ordering on the sheaf Laplacian with non-trivial structure has differential entries $|v_1(a_{i+1}) - v_1(a_i)|$ that are large on chain-extending edges and small on redundant edges (because redundant edges lie within already-connected components of the chain). The joint policy and Fiedler tie-break consistently prefer chain-extending edges; the worst-case path length is $n + r$. $\square$

Remark 4 (Relation to v2’s flawed proof). The v2 proof sketch of Theorem 3(a) implicitly assumed all **rw** candidates would apply; in fact, the kernel rejects rewrites whose terms are not in the goal, and these rejected attempts consume expansions. A14 fixes this by filtering them before they reach the kernel, restoring the $O(n)$ bound as a property of the implemented system.

Remark 5 (Rarity acknowledged). Pure equality-chain states are rare in mathlib (plausibly $< 5\%$ of the held-out slice). Theorem 3 contributes a *mechanistic* result, not a headline pass-rate result. The empirical correlation between λ_1 and expansion count on detected chain states is reported as a *sanity-check experiment* in §7 (paper), not as a contribution metric.

Remark 6 (Fiedler still adds value beyond A14). On pure linear chains, A14 alone suffices to give the $O(n)$ bound; the Fiedler vector becomes a tie-break. On mixed states (chain + irrelevant equalities passing the textual filter), Fiedler’s spectral contrast is what discriminates chain edges from irrelevant ones.

8 Threats to validity

8.1 Threat (i): M5/M3 search-tree overlap

If Mode 2 top- k of M5 and M3 overlap heavily, search trees converge. Signal: $\Delta(B) \rightarrow 0$; Jaccard > 0.9 ; tree overlap > 0.85 . Contingency: emphasise qualitative + Theorem 3 mechanistic evidence.

8.2 Threat (ii): Mode 2 head fails to learn (updated under A10)

The Mode 2 head’s frozen-encoder bottleneck of v2 is partially addressed in v3 by A10 (top-2 encoder layers unfrozen). If even with A10 the held-out within-family top-1 accuracy of Mode 2 is at floor (uniform), the v3 contingency tightens:

- *Confirming signal*: Mode 2 top-1 $\approx 1/|G_{\text{cand}}(P, F)|$ on validation.
- *Contingency (graduated)*:
 1. Increase unfreeze depth to top-3 (+5 h training).
 2. If still floor, unfreeze full encoder (+15 h training).
 3. If still floor, report Mode 2 ineffective; revert to Mode-1-only.

8.3 Threat (iii): coverage rate too low even with B2 retrieval

BM25 is sparse, surface-form. For mathlib non-local proofs that require semantic retrieval, BM25 will miss the correct lemma.

- *Signal*: coverage rate (with B2) $< 60\%$ on the held-out slice *before* applying the B1 secondary filter.
- *Contingency*: report the B1-filtered slice’s coverage as the achievable upper bound. The headline hypothesis is tested on the filtered slice, not on arbitrary mathlib.

8.4 Threat (iv): gap too small to detect

Three seeds (C7) help. CI overlap with zero remains possible. Report effect sizes regardless.

8.5 Threat (v): LeanDojo / mathlib brittleness

Same as v2: log all LeanDojo failures with reason codes; re-run flagged theorems once; report.

8.6 Threat (vi): equality-chain subcase contributes little

Even sharp advantage on the chain slice may not move the headline. Mitigation: report the chain-slice separately with the λ_1 -vs-expansions correlation.

8.7 Threat (vii): hybrid dominates leakage-safe

If $G_{\text{cand}}^{\text{hyb}}$ outperforms G_{cand} substantially, this suggests retrieval candidates are insufficient and the LM is doing the real work.

8.8 Threat (viii): A14 filter overfits to pure chains (new in v3)

If the detection rule fires falsely (or fails to fire on near-chain states), the search may proceed sub-optimally. Mitigation: report the detection rule’s precision/recall on a 1000-state validation sample.

8.9 Threat (ix): A12 compiler-feedback signal too weak (new in v3)

With 500 states for A12, the negative-signal coverage is small.

- *Signal*: held-out validation accuracy with A12 $\approx$ accuracy with A11 only.
- *Contingency*: report A12 as “no measurable effect”; document that full DPO would require ~ 30 –50 h more compute. Future-work-flagged.

9 Paper outline

Target: 14–18 pages; cap 20.

§1 Introduction (2 pages). Scope statement on page 1:

This paper couples a topological-deep-learning representation of Lean 4 proof states (Role-Labelled Incidence Complexes, RLIC-1) into a length-normalised best-first proof search with two-stage neural policy: family-level Mode 1 plus instance-level Mode 2. Candidate generation is leakage-safe under Condition 3.6 of RLIC-1 and is augmented with a 3.6-compliant retrieval shortlist. At fixed parameter scale (~ 36 M) and on a held-out mathlib slice constructed to be within the candidate generator’s expressive reach, the full RLIC architecture (M5) outperforms a strong role-aware hyperedge baseline (M3) at matched parameters and search compute. This is a methodology / inductive-bias study, not a SOTA chase.

§2 Related work (1.5 pages). AlphaProof, DeepSeek-Prover-V1.5/V2, Kimina-Prover, BFS-Prover (our algorithmic template), Goedel-Prover, ReProver / LeanDojo, Graph2Tac, HILBERT.

§3 The coupled system (2.5 pages). Mode 1 / Mode 2; G_{cand} ; equality-chain subcase; retrieval shortlist; Mode 2 training (A10, A11, A12).

§4 The search procedure and principal hypothesis (1.5 pages).

§5 Mathematical content (2.5 pages). Theorems 1, 2, 3 (reformulated under A14).

§6 Experimental protocol (2 pages).

§7 Results (3 pages). Held-out mathlib results (primary); miniF2F-100 diagnostic; Mode 2 top-1 and coverage; equality-chain slice with λ_1 correlation; A12 ablation.

§8 Qualitative evidence (1 page). IHES card.

§9 Limitations and threats (1.5 pages).

§10 Conclusion (0.5 pages). Re-state scope; forward-point to full DPO, retrieval-augmented simp candidates, MCTS at larger scale.

Total: 17.5 pages.

10 Estimated compute budget

10.1 Per-expansion cost

~ 1.2 s wall per expansion (LeanDojo dominated). Neural forward including Mode 2 batch ~ 50 ms. With 5-minute per-theorem wall cap, average per-theorem cost ~ 2.5 –3 min at $B = 256$.

10.2 v3 budgeted runs

Run	Hours	Notes
Mode 2 training Phase 1, A10 partial unfreeze (3 architectures)	18	6 h each
Mode 2 training Phase 2, A12 compiler-feedback (data + 3 architectures)	8.5	4 h data + 4.5 h training
Retrieval index build (B2, BM25)	1	one-time
Small LM (Pythia-160M LoRA) training	5	one-time
Hyperparameter sweep over α_{len} (50-theorem dev split)	6	one round per system class
Primary: held-out mathlib 500 theorems @ B = 256		
M5+M2 $\times 3$ seeds	60	20 h/run $\times 3$
M3+M2 $\times 3$ seeds	60	
M3++M2 $\times 1$ seed	20	grading-isolation control
Small LM $\times 1$ seed	20	compute-parity text baseline
No-policy $\times 1$ run	11	fast failures
Diagnostic: miniF2F-100 @ B = 256		
M5+M2 $\times 1$ seed	5	
M3+M2 $\times 1$ seed	5	
M3++M2 $\times 1$ seed	5	
Small LM $\times 1$ seed	5	
M5+Mode-1-only $\times 1$ seed	5	Mode 2 ablation
No-policy $\times 1$ run	3	
Qualitative IHES card (10 thms $\times 2$ systems $\times B = 512$)	4	
Sanity checks (Appendix A)	3	
Total	244.5	

Contingency: 5.5 hours. Tight but workable. Preallocated to LeanDojo retries, qualitative-evidence supplements, and A12 data-collection slippage.

10.3 What v3 has added relative to v2

A10 (partial encoder unfreeze): +6 h. A12 (compiler-feedback phase): +8.5 h. B2 (retrieval index): +1 h. A11, A14: 0 h.

10.4 What v3 has cut relative to v2

miniF2F diagnostic: $244 \rightarrow 100$ theorems, saves ~ 20 h. B1 shorter proofs (≤ 15 instead of ≤ 30): saves ~ 5 h. Net change ≈ -10 h. v3 budget = 244.5 h vs v2’s 243 h. Approximately budget-neutral.

10.5 Calendar time

244.5 h on a single 4060 ≈ 10.2 days continuous. With realistic interruptions, 4–5 calendar weeks.

10.6 What we are not doing

- No third seed on M3+, LM, or auxiliary systems.
- No held-out mathlib at 1000 theorems (kept at 500).
- No `novel_premises` split.
- No MCTS comparison.
- No Rocq.
- No ReProver head-to-head (cite published numbers).
- No full BFS-Prover-style DPO with compiler feedback at scale (A12 is the cheap analogue; full DPO is future work).
- No retrieval-augmented `simp` candidates (only `apply` in v3).
- No unfrozen full encoder during Mode 2 training (Threat 8.2 contingency only).

A Sanity checks before any experiment runs

1. Canonicalise miniF2F-test against RLIC-1 train; assert zero collision.
2. Build the held-out mathlib slice with B1 filter; report actual N (target 500, accept ≥ 300); report coverage rate of human tactics under $G_{\text{cand}} + \text{B2}$ retrieval.
3. Build the BM25 retrieval index (B2); assert held-out slice’s theorems are excluded; report index size.
4. Run Algorithm 4 no-leakage probes on 1000 states from each set; assert $\geq 99\%$ pass-rate. Additional probe: B2 retrieval is α -renaming invariant.
5. Re-confirm M5/M3 Mode-1 top-1 on the RLIC-1 validation set.
6. Time 20 random `run_tac` calls with M5 + Mode 2 forward in the loop; assert per-expansion cost ≤ 2 s.

7. **(A14)** Equality-chain detection sanity on a 1000-state validation sample; report fraction detected; compute $\lambda_1(L)$ for detected states; assert numerical well-conditioning (condition number $< 10^8$); manually verify chain structure on 5 randomly chosen detected states.
8. **(A10)** Mode 2 training sanity: after Phase 1 with A10 partial unfreeze, assert held-out within-family top-1 $> 30\%$; report per-family Mode 2 accuracy; report coverage rate of $G_{\text{cand}} + \text{B2}$.
9. **(A12)** Compiler-feedback phase sanity: after data collection, report distribution of accepted/rejected/timeout candidates per family; assert at least 40% of states have both accepted and rejected candidates.
10. **(B2)** Retrieval sanity on 50 random training states: manually inspect top-8 BM25 retrievals; assert that for at least 30% of states, the human’s applied lemma is in the top- K for some $K \leq 16$.
11. Run a 10-theorem trial search with M5+M2 at $B = 64$; assert at least one proof is returned and Theorem 1 replay verification passes.
12. Run the hyperparameter sweep over α_{len} on the 50-theorem development set; record chosen value; commit.

B Pre-registered tables and plots

Table 1: Held-out mathlib pass-rates at $B = 256$. Columns: System; trainable params; leakage-safe ($G_{\text{cand}} + \text{B2}$) pass-rate; hybrid pass-rate; 95% CI; coverage rate. Rows: M5+M2, M3+M2, M3++M2, LM, No-policy.

Figure 1: Pass-rate vs. budget B (held-out mathlib). X-axis B ; Y-axis pass-rate. Lines: M5+M2, M3+M2, M3++M2, LM, No-policy with seed bands. $\Delta(B)$ gap on secondary axis.

Table 2: Sliced held-out-mathlib pass-rates at $B = 256$. Slices: rewrite-heavy, multi-arg-app, induction, residual, equality-chain (new under A14). Per-system columns.

Table 3: Mode 2 top-1 accuracy and G_{cand} coverage rate by system. Threat 8.2 diagnostic.

Table 4: miniF2F-100 diagnostic. All systems including Mode-1-only ablation at $B = 256$.

Table 5: IHES qualitative card. 5–10 rows.

Table 6: Equality-chain mechanistic check. Per chain-state: $\lambda_1(L)$, M5+M2 expansion count, M3+M2 expansion count, A14 filter pass/fail rate. Scatter plot $\log(\text{expansions})$ vs. $\log(1/\lambda_1)$.

Table 7 (new in v3): A12 ablation — Phase 1 only vs. Phase 1+2. M5+M2 pass-rate on held-out mathlib at $B = 256$ with and without the compiler-feedback phase. Tests Threat 8.9.

Table 8: Threats-to-validity materialisation. Nine rows (i)–(ix). Filled after experiments.

C v1 → v2 Changelog (retained for traceability)

Four amendments accepted from the first critique exchange, plus four tightly-coupled changes:

#	Change	Why
C1	Mode 2 promoted to first-class architecture.	Mode-1-only left within-family ordering at the mercy of syntactic position.
C2	Value head dropped from priority function ($\beta = 0$).	Goal-swap-trained V predicted to fail on structurally coherent real failures. BFS-Prover demonstrated BFS without V is competitive.
C3	Held-out mathlib slice promoted to primary; miniF2F-test demoted to diagnostic.	miniF2F has frontier SOTA at 86–99%; 36 M absolute pass-rates there are uninformative.
C4	Theorem 7.4 reformulated; G_{cand} modified on equality-chain subcase to consult sheaf Laplacian.	v1 statement was vacuous: G_{cand} ordered by syntactic position, spectral data never reached search order.
C5	Priority function switched to length-normalisation.	Follows from C2 and BFS-Prover’s practice.
C6	ReProver dropped as head-to-head baseline; cited only as scale-context.	2023 system; anachronistic.
C7	Third seed restored on M5 and M3.	2-seed power insufficient for sub-2pp effect size.
C8	Scope statement reframed: methodology study, not SOTA chase.	Honest positioning at fixed parameter scale.

Closing (v3)

Eight amendments across two critique rounds have made the design materially stronger:

Round 1 (v1 → v2). Mode 2 promoted; value head removed; held-out mathlib slice elevated; Theorem 3 reformulated. The principal hypothesis is now tested on a no-frontier-SOTA slice with within-family neural ordering and length-normalised priority following BFS-Prover.

Round 2 (v2 → v3). Mode 2 head capacity strengthened via partial encoder unfreeze (A10); training augmented with hard-negatives reframing (A11) and a compiler-feedback phase (A12) as a cheap DPO-substitute; candidate generator augmented with 3.6-compliant BM25 retrieval (B2) and an equality-chain textual-presence filter (A14, also closing a genuine flaw in v2’s Theorem 3 proof); held-out mathlib slice tightened to within G_{cand} ’s expressive reach (B1); diagnostic miniF2F reduced to 100 theorems to fit the compute budget.

What this document does *not* promise:

- Competitiveness with BFS-Prover, DeepSeek-Prover-V2, Kimina-Prover, or any frontier LLM-based prover on absolute pass rates.
- A breakthrough on miniF2F-test.
- That topological inductive biases will scale to 7B / 32B parameters.
- That Mode 2 will learn well even with A10 (Threat 8.2 contingency).
- That A12's compiler-feedback phase will materially improve over A11 alone (Threat 8.9 contingency).
- That BM25 retrieval has enough semantic depth (Threat 8.3 contingency).

What it does promise:

- A pre-registered, leakage-safe, compute-parity end-to-end evaluation of the RLIC architecture against parameter-matched hyperedge baselines, on a slice explicitly constructed to be within the candidate generator's expressive reach.
- A rigorous theorem connecting the cellular sheaf Laplacian on equality-chain proof states to search-complexity bounds in the actually implemented search algorithm.
- A small but real DPO-substitute via compiler-feedback hard negatives.
- Honest contribution claims and contingencies for a methodology study at fixed scale.

When this document is reviewed and confirmed internally consistent, move to Prompt 2 (implementation). Any change during implementation is added as a dated amendment to this file.